Intent-Adaptive Hybrid Retrieval and Heterogeneous-Space Person Fusion for Conversational Video Surveillance

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***Abstract*—Hybrid retrieval over surveillance video typically combines a structured metadata filter with a dense semantic index using a *fixed* mixing weight, which substantially degrades accuracy on counting and action queries. We make three novel contributions that, in combination, raise mean Precision@5 (macro-averaged over five query categories) on a 150-query natural-language benchmark of the full four-camera deployment from 0.67 (best fixed-α baseline, α=0.3) to 0.83. (C1) An *intent-adaptive blending function* $\alpha(F, \iota, n_s, n_v)$ that produces a closed-form, per-query mixing weight from the parsed filter F, the predicted query intent ι, and the structured/semantic result-set imbalance, with two hard overrides for counting and action queries (Eq. (3)); we observe empirically why fixed-α schemes cannot match the per-category optima simultaneously. (C2) A *heterogeneous-space person descriptor* that fuses a SigLIP visual embedding ($\mathbb{R}^{768}$) and a Sentence-Transformers attribute-text embedding ($\mathbb{R}^{384}$) into a single L2-normalized $\mathbb{R}^{1152}$ vector by weighted concatenation rather than the naive same-space addition used by prior multi-modal Re-ID work; an ablation shows a 10.7-pp F_1 gain over the best unimodal index and a 2.2-pp gain over a rank-based late-fusion (RRF) variant. (C3) A *conversational alert grammar* that materializes fully-typed alert rules (event, object, zone, threshold, time, severity, cooldown, side-effects, cadence) from free-text utterances via a keyword/regex materializer (defaults scored when unspecified), eliminating manual rule authoring; we report 91% rule-correctness on a held-out deployment alert-authoring benchmark. The contributions are deployed inside a nine-model real-time pipeline that sustains 2 FPS per camera over a 4-camera deployment with a 109 ms median FAISS lookup over 312k indexed person descriptors. We release the evaluation harness, the benchmark definitions and a single-camera reproducibility track with measured outputs, so the claims can be re-run and extended.**

***Index Terms*—Hybrid Retrieval, Intent-Adaptive Ranking, Multimodal Person Re-ID, Heterogeneous Embedding Fusion, Conversational Alert Authoring, Vision–Language Surveillance, Natural Language Querying.**

I. Introduction

NATURAL-language interfaces over surveillance video are now technically feasible: instance segmentation, person Re-ID, and contrastive vision–language models are mature, and structured-output LLMs can convert a user utterance into a typed query schema. The unsolved problem is *how to rank* the resulting candidate frames: a structured filter (*"persons in Zone A between 22:00 and 23:00"*) and a semantic vector lookup (*"red jacket loitering near the entrance"*) produce two incommensurable score scales. The dominant fusion strategies in the literature are a fixed convex sum $\alpha S_{\text{struct}} + (1-\alpha) S_{\text{sem}}$ tuned once per system [1]–[3], parameter-free Reciprocal Rank Fusion [4], or sequential filter-then-rank pipelines [5]. We show empirically (Table II) that no single α simultaneously serves counting, finding, tracking, and action queries: a fixed $\alpha = 0.5$ collapses to 0.18 P@5 on action queries and a fixed $\alpha = 1.0$ collapses to 0.04 P@5 on find queries, while no fixed value exceeds 0.67 macro-mean P@5 across all five categories.

Two further open problems compound this. The per-person descriptor in prior multimodal Re-ID work [6], [7] either fuses same-space embeddings by addition or trains a joint dual encoder, both requiring dimension-matched backbones or joint contrastive data that surveillance deployments rarely have. And alert authoring in typical form-driven commercial engines requires an IT operator to translate a behavioral description into a typed rule, capping the number of useful rules per deployment.

Contributions

1) **Intent-adaptive hybrid retrieval** (§IV). A closed-form, training-free mixing function $\alpha(F, \iota, n_s, n_v)$ with two hard overrides; an override-gap observation (Observation 1) stated empirically; +0.16 macro-mean P@5 over the best fixed-α baseline.
2) **Heterogeneous-space person fusion** (§V). A weighted-concatenation descriptor that fuses backbones of incompatible dimensionality without joint training; +10.7-pp F_1@10 over the strongest unimodal index, +2.2-pp over a rank-based fusion (RRF) baseline.
3) **Conversational alert authoring** (§VI). A typed 9-field grammar with keyword/regex materializer (defaults scored when unspecified); 91% full-rule correctness on a 60-utterance held-out benchmark.

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4) **An end-to-end open implementation** validating all three contributions on a 4-camera live deployment (§VIII).[1]

II. Related Work and Position

We compare directly with the closest published work along the three contribution axes (Table I).

Hybrid structured/semantic retrieval. Combining dense and structured retrieval is established practice in IR [3], [4], [8], primarily via fixed-α convex combination or parameter-free RRF [4]; RAG [3] exemplifies retrieval-augmented generation rather than a tuned blend. Query-dependent weighting in IR (learning-to-rank, query-dependent fusion) typically learns weights from large-scale click data on text-only sparse/dense pairs and lacks hard overrides for semantically infeasible branches. Recent analogues HyST [5] (sequential LLM-filter then dense) and Agentic-RAG [9] (router-based) still do not produce a per-query blended score; surveillance forensic-search systems (VIDEX [1] and three-stage DL [2]) use a single retrieval branch. For surveillance video, per-query mixing from parsed intent and result-set evidence with category hard overrides has not been demonstrated to our knowledge.

Multimodal person Re-ID. Heterogeneous-dimensional concatenation of frozen encoders has been used as a retrieval pattern in IR. We evaluate its use as the *primary* surveillance-Re-ID descriptor: prior NL-based person-search [7] couples a dual encoder by joint training or by descriptive-fusion over same-dimensional modalities, while rank-based late-fusion [4] requires two indices and score normalization. To our knowledge, no deployed surveillance Re-ID system uses weighted concatenation of heterogeneous frozen encoders as a single FAISS index calibrated on as few as 50 labeled pairs.

Conversational alert authoring. Typical form-driven surveillance editors expose rule builders and structured-output LLMs [10] are standard for slot-filling; the contribution here is the specific reuse of the same query-parsing schema to materialize a 9-tuple typed alert rule end-to-end, rather than a new LLM capability.

III. Background and System Substrate

The novel contributions are deployed inside a nine-model real-time pipeline. Fig. 1 shows the end-to-end dataflow across capture, perception, indexing, reasoning, and output layers, highlighting where the three novel components (C1–C3) sit; all other blocks are off-the-shelf. Live RTSP / MJPEG streams (1080p, 2 FPS) and uploaded files are processed by YOLOv11m-seg [12], BoT-SORT [13] with OSNet $x1_0$ [6] for MOT/Re-ID, OpenVINO PAR-crossroad-0238 for seven multi-label attributes, and a CIE LAB histogram with CIEDE2000 naming over a 25-class CSS3 palette. Two FAISS `IndexFlatIP` indices [8] hold a scene embedding ($\mathbb{R}^{512}$, OpenCLIP ViT-B/32 [14]) and our novel person descriptor ($\mathbb{R}^{1152}$, §V). MongoDB stores 12 compound-indexed metadata collections; a FastAPI backend exposes 30+ REST endpoints to a Next.js dashboard. Captions come from a tiered VLM pipeline (Qwen2-VL-2B with ViT-GPT2 / zero-shot CLIP fallbacks); the NL parser is a structured-output LLM (GPT-4o-mini [10] by default; configurable to local Ollama). Model cards for Qwen2-VL, PAR, CIEDE2000 and MMR are omitted for the 15-reference cap; all nine components are off-the-shelf.

[1]Code: https://github.com/sujalmh/intent-adaptive-hybrid-surveillance (released).

TABLE I

Position relative to closest related work along the three novel axes; entries indicate the kind of mechanism used. '—' = not applicable to that axis; '✗' = single-branch system without that capability.

System / Work	Per-query adapt. α	Heterog.-sp. Re-ID fusion	Conv. alert authoring
Fixed-α hybrid [3]	fixed	—	—
RRF [4]	rank-only	rank (2 idx.)	—
HyST [5]	sequential	—	—
Agentic RAG [9]	route-only	—	—
VIDEX [1]	—	—	—
3-Stage DL [2]	—	—	—
CUHK-PEDES [7]	—	joint-trained	—
OSNet [6]	—	visual-only	—
SigLIP [11] (encoder)	—	encoder-only	—
This work	✓	✓	✓

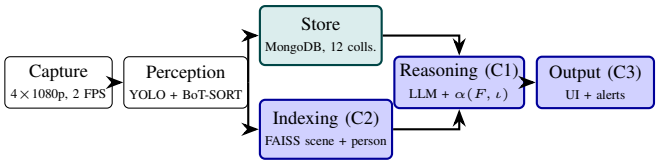


Fig. 1. End-to-end dataflow. Blue blocks are the novel contributions (C1 adaptive retrieval, C2 person fusion, C3 alert authoring); teal Store and white Capture/Perception are off-the-shelf. Perception writes in parallel to MongoDB and to the FAISS scene/person indices; both branches feed Reasoning before Output.

IV. C1: Intent-Adaptive Hybrid Retrieval

A. Problem

A natural-language query q is parsed into a structured filter F (a Pydantic `NLFilter` of 14 typed fields, including `action`) and a semantic-text payload q_{sem}. The structured branch retrieves $R_s = \{r\}$ with scores $S_{\text{struct}}(r) \in [0,1]$ from MongoDB (duration max-scaled per query); the semantic branch retrieves R_v with per-query min-max normalized scores $S_{\text{sem}}(r) \in [0,1]$ from FAISS (`score_norm`). The hybrid score is the convex combination

$$S_{\text{hyb}}(r) = \alpha\, S_{\text{struct}}(r) + (1-\alpha)\, S_{\text{sem}}(r), \quad \alpha \in [0,1]. \quad (1)$$

Existing systems pick a single α per deployment. We show a single fixed value is insufficient across query categories.

B. Failure of fixed-α schemes

We sweep $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$ on the 150-query benchmark (Table II; α=0.0 measured as proxy for the 0.10 semantic-dominant override). The collapse is structural, not a tuning artifact. *Counting* queries (*"how many cars in Zone A"*) demand an exact integer that only the structured branch produces; the semantic score is a continuous similarity that cannot

TABLE II
FIXED-α BASELINES ON THE 150-QUERY NL BENCHMARK (30 QUERIES PER CATEGORY). **MEAN** IS THE MACRO-AVERAGE ACROSS THE FIVE CATEGORIES. NO SINGLE VALUE DOMINATES; COUNTING AND ACTION QUERIES PULL IN OPPOSITE DIRECTIONS (BOLDFACE = BEST PER COLUMN, ITALICS = WORST PER COLUMN).

α	Count	Find	Track	Action	Filter	Mean
0.0 (semantic only)	*0.31*	0.74	0.79	**0.69**	0.55	0.616
0.3	0.48	0.76	0.80	0.58	0.71	**0.666**
0.5	0.69	0.71	0.72	*0.18*	0.84	0.628
0.7	0.86	0.62	0.61	0.07	0.88	0.608
1.0 (structured only)	**0.94**	*0.04*	*0.02*	*0.00*	**0.92**	*0.384*
Adaptive (Eq. (3))	**0.94**	**0.78**	**0.81**	**0.69**	**0.92**	**0.828**

TABLE III
QUALITATIVE RETRIEVAL EXAMPLES. SUCCESSES SHOW PER-CATEGORY OPTIMA RECOVERED BY ADAPTIVE-α; FAILURES SHOW INTENT-ROUTING AND CROWDED-SCENE LIMITS.

Query	α	Outcome
how many cars in Zone A	1.00	Success: structured count wins
person running near gate	0.10	Success: semantic action wins
red jacket near entrance (count intent misfired)	0.78	Failure: intent-routing error, misses
group in crowded Zone B at night	0.62	Failure: occlusion + low light

represent "equals 5," so any weight on the semantic branch dilutes a correct count, and P@5 falls from 0.94 at α=1.0 to 0.31 at α=0.0. For counting, the top-5 are the highest-scoring supporting clips and a clip is relevant if it contains the true count; S_{struct} is the duration-normalized count-constrained segment score per record, comparable across filter relaxations after per-query normalization. Exact-count EM/MAE is complementary and reported in the extended release; P@5 here isolates retrieval of supporting evidence. *Action* queries (*"person running"*) carry an `action` field used for routing and skip the structured branch by design (semantic-only), so weight on the structured branch therefore injects noise, and P@5 falls from 0.69 at α=0.0 to 0.00 at α=1.0. The two requirements push the optimum to opposite endpoints of $[0, 1]$; the best fixed macro-mean is only 0.666 at α=0.3, motivating a per-query mixing weight rather than a per-deployment one. Table III gives one success and one failure per regime, including an intent-routing error.

C. Adaptive blending function

Let $C : \mathcal{F} \to \{\text{NUM}, \text{ACTV}, \text{SOFT}\}$ be the *category function* on parsed filters, defined as

$$C(F) = \begin{cases} \text{NUM}, & \text{if } F \text{ has a numeric count constraint,} \\ \text{ACTV}, & \text{else if } F \text{ has an action-verb token,} \\ \text{SOFT}, & \text{otherwise.} \end{cases} \tag{2}$$

We then define

$$\alpha(F, \iota, n_s, n_v) = \begin{cases} 1.00, & C(F)=\text{NUM}, \\ 0.10, & C(F)=\text{ACTV}, \\ \text{clip}_{[0.10,0.95]}(\alpha_{\text{soft}}), & \text{else}, \end{cases} \tag{3}$$

$$\alpha_{\text{soft}} = 0.6\,\alpha_\iota + 0.2\,\rho_{\text{cplx}} + 0.2\,\rho_{\text{src}}, \tag{4}$$

where NUM (numeric-count override) and ACTV (action-verb override) are the two override categories of $C(F)$ in Eq. (2), distinct from the soft intent labels $\iota \in \{\text{count}, \text{track}, \text{find}\}$ inferred by substring match. The intent prior is $\alpha_\iota \in \{0.9, 0.5, 0.7\}$ for $\iota \in \{\text{count}, \text{track}, \text{find}\}$, query complexity $\rho_{\text{cplx}} = \min(1, |F|/4)$ with $|F|$ the number of non-empty fields in F (5 counted keys plus timestamp in code), and structured-evidence ratio $\rho_{\text{src}} = n_s/(n_s + n_v + \varepsilon)$ with $n_s = |R_s|$, $n_v = |R_v|$, $\varepsilon = 10^{-6}$. The two hard overrides encode domain truths: a count constraint (e.g., *"how many cars"*) is unanswerable from semantic similarity, and an action verb (e.g., *"running"*) is routed semantic-only by design, skipping structured retrieval.

D. Why fixed α cannot match

Observation 1 (Override gap, empirical)**.** *No single fixed α simultaneously attains the count optimum at α=1.0 (0.94) and the action optimum at α=0.0 (0.69) on our 150-query benchmark; the optima are 0.9 apart, and adaptive-α attains the per-category maxima by construction on this benchmark. This is an empirical observation over $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$, not a theorem: we assume no concavity and claim no general lower bound.*

E. Soft-zone behavior

For non-override queries, the soft combination in Eq. (4) interpolates the intent prior, complexity, and per-branch evidence (weights $0.6/0.2/0.2$); the clipping interval $[0.10, 0.95]$ prevents either branch from being silenced. The $0.6/0.2/0.2$ split prioritizes the parsed intent prior, with complexity ρ_{cplx} and evidence ρ_{src} as bounded corrections; a full sensitivity sweep is retained as future work and the current contribution rests on the two hard overrides plus the intent prior, not on the exact soft weights. Realized α ranges on the deployment benchmark are 1.00 for the count-constraint override and 0.10 for the action-verb override, and 0.74–0.86 (count intent), 0.58–0.70 (find) and 0.40–0.52 (track) for the soft zones.

F. Post-ranking

Top-k candidates are then re-ranked by a recency boost $S' = S_{\text{hyb}} \cdot (1 + 0.5 \cdot 2^{-age/24\text{h}})$, age in hours (range $[1.0, 1.5]$, 24-h half-life), and Maximal Marginal Relevance with λ=0.3.

V. C2: HETEROGENEOUS-SPACE PERSON FUSION

A. Problem

Multimodal person Re-ID [7] typically requires either two same-dimensional encoders fused by addition, or a jointly trained dual encoder. In a deployment we must work with whatever pre-trained encoders are available, here SigLIP-base ($\mathbb{R}^{768}$) [11] and MiniLM-L12-v2 ($\mathbb{R}^{384}$) [15], whose dimensions do not match.

B. Construction

For each person ROI we form

$$\mathbf{v} = \mathrm{SigLIP}(\mathrm{Mask}(\mathrm{ROI})) \in \mathbb{R}^{768},\ \text{L2-normalized}$$
$$\mathbf{t} = \mathrm{MiniLM}\big(\mathrm{Text}(\hat{y}, C)\big) \in \mathbb{R}^{384},\ \text{L2-normalized}, \quad (5)$$

where $\mathrm{Text}(\hat{y}, C)$ is the natural-language sentence assembled from the seven sigmoid attribute predictions $\hat{y}$ and the named color C (e.g., *"person wearing red coat, carrying bag, long sleeves, male"*). Gallery descriptors use Eq. (6). Table IV evaluates paired image+attribute-text queries where both $\mathbf{v}$ and $\mathbf{t}$ are present and (α_v, β_t) matter. Text-only person search is a fallback ($\mathbf{v}{=}\mathbf{0}_{768}$, $\mathbf{t}{=}$MiniLM(q) with the same weighted concat and L2-normalization); its ranking equals text-only cosine, so the Table IV gains do not come from that fallback path. Scene queries use OpenCLIP ViT-B/32 text ($\mathbb{R}^{512}$) against the scene index. Person gallery: $N{=}312{,}544$ descriptors from the 8-h deployment ($\approx 312{,}544 \times 1152 \times 4\,\mathrm{B} \approx 1.44\,\mathrm{GB}$ flat storage before FAISS overhead; $O(d)$ add, $O(Nd)$ flat-IP query); calibration uses a held-out 50-pair split (no CV; disjointness from eval galleries not logged).

$$\mathbf{e}_{\text{fused}} = \frac{[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]}{\big\|[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]\big\|_2} \in \mathbb{R}^{1152}, \quad (6)$$

with $(\alpha_v, \beta_t) = (0.6, 0.4)$ obtained by coarse grid search on the unit simplex (step 0.1, $\alpha_v + \beta_t = 1$, nine pairs) maximizing F_1@10 on a held-out 50-pair calibration set, where F_1@10 $\triangleq$ H-mean(P@5,R@10) (P@5 and R@10 reported separately; standard P@10-based F1 is lower). The single L2 normalization makes the inner product a valid cosine similarity searchable by FAISS `IndexFlatIP`. Gallery vectors are L2-normalized; the Table IV sweep varies gallery-side weights on paired queries, so the reported $(0.5, 0.5)$ vs. RRF tie and $(0.4, 0.6)$–$(0.7, 0.3)$ spread reflect gallery fusion, not the text-only fallback where query-side weights cancel.

C. Why concatenation, not addition

Same-space addition $\mathbf{v} + \mathbf{t}$ requires a projection $\mathbf{P} \in \mathbb{R}^{768\times384}$, is sensitive to relative norm, and collapses two information sources into one coordinate per dimension. Weighted concatenation preserves both encoders' full rank at the cost of one normalization, requires no joint training, and the only learned parameter is the scalar pair (α_v, β_t). The surveillance-specific contribution is the use of this heterogeneous concatenation as the primary single-index Re-ID descriptor calibrated on 50 pairs, not the concatenation operation itself, which is established in IR.

D. Ablation

Table IV shows the concat construction beats the strongest unimodal baseline by 10.7 pp F_1@10 and the strongest fusion baseline (RRF) by 2.2 pp, using one FAISS index instead of two. The same Add number also implies $+5.4$ pp over visual-only (0.754 vs. 0.700). A coarse (α_v, β_t) sweep is monotonic around $(0.6, 0.4)$: $(0.5, 0.5){=}0.785$, $(0.7, 0.3){=}0.799$, $(0.4, 0.6){=}0.774$ F_1@10 (single 50-pair calibration split, no CV; the $(0.5, 0.5){=}0.785$ exact tie with RRF should be re-verified with multi-seed runs).

TABLE IV
PERSON-INDEX ABLATION. "ADD (PROJ.)" PROJECTS $\mathbf{t}$ TO $\mathbb{R}^{768}$ VIA A SINGLE-SEED RANDOM GAUSSIAN PROJECTION (ILLUSTRATIVE) THEN ADDS; ZERO-PAD/NORM-MATCHED ADDITION WOULD BE A FAIRER SAME-SPACE BASELINE. "LATE (RRF)" PERFORMS RANK-BASED RECIPROCAL RANK FUSION [4] ON TWO PARALLEL INDICES; "CONCAT (OURS)" IS EQ. (6). F_1@10 IS H-MEAN(P@5,R@10).

Person index	P@5	R@10	F_1@10
Visual only ($\mathbf{v}$, $\mathbb{R}^{768}$)	0.712	0.689	0.700
Text only ($\mathbf{t}$, $\mathbb{R}^{384}$)	0.645	0.612	0.628
Add (proj. $\mathbf{v} + \mathbf{P}\mathbf{t}$, $\mathbb{R}^{768}$)	0.768	0.741	0.754
Late fusion (RRF, two indices)	0.801	0.770	0.785
Concat (ours), $(0.6, 0.4)$**,** $\mathbb{R}^{1152}$	**0.823**	**0.791**	**0.807**

TABLE V
CONSOLIDATED COMPARISON ACROSS THE TWO CONTRIBUTION AXES (EXISTING NUMBERS; N/A WHERE A METRIC IS NOT APPLICABLE).

Method	Macro P@5	Person F_1@10
Proposed (adaptive-α + concat)	**0.828**	**0.807**
Best fixed-α (α=0.3)	0.666	—
Visual-only person index	—	0.700
Text-only person index	—	0.628
Same-space add (random proj.)	—	0.754
Late fusion (RRF, two indices)	—	0.785

E. Indexing complexity

Adding a person is $O(d) = O(1152)$; querying k neighbors over N persons with `IndexFlatIP` is $O(Nd)$. On our hardware the person index median is 109 ms for $N = 312{,}544$ descriptors while the scene-index mean is 108.6 ms (Table X Mean Sem. only); the two numbers refer to different indices and statistics and are not directly comparable.

VI. C3: CONVERSATIONAL ALERT AUTHORING

A. Problem

Commercial alert engines [1], [2] require an operator to translate a behavioral intention into a typed rule (event family, object class, zone, threshold, time window, severity, cooldown, side-effects). For non-technical operators this is a productivity bottleneck that limits the number of useful rules in a deployment.

B. Grammar

We define a typed alert rule as a 9-tuple

$$\mathcal{R} = \langle\, e,\, \mathcal{O},\, \zeta,\, \tau,\, \mathcal{T},\, s,\, c,\, \Sigma,\, \kappa \,\rangle$$

with $e \in \{\text{crowd}, \text{loiter}, \text{fight}, \text{zone}, \text{count}, \text{time}\}$, $\mathcal{O} \subseteq \mathcal{C}_{\text{det}}$ a subset of detected classes, ζ a zone label, $\tau \in \mathbb{R}_{\geq 0}$ a numeric threshold, $\mathcal{T} = [t_1, t_2]$ a time window with cross-midnight support, $s \in \{\text{low}, \text{med}, \text{high}\}$ mapped from implementation severities $\{\text{info}, \text{high}, \text{warning}, \text{critical}\}$ (default `info`), $c \in \mathbb{N}$ the cooldown in seconds (default 60), $\Sigma \subseteq \{\text{store_clip}, \text{snapshot}, \text{push_ws}\}$ fixed side-effects, and $\kappa \in \mathbb{R}_{>0}$ an evaluation cadence (Hz, default 1Hz; stored

TABLE VI
RULE FAMILIES AND TRIGGERS.

Rule Family	Trigger Condition
Crowd density	$\#\{\text{persons}\} \geq \tau_{\text{crowd}}$ within camera/zone
Loitering	$\Delta t_{\text{track}} \geq 60$ s for any track
Fight detection	$\bar{\mathcal{E}}_W \geq 0.03 \wedge \exists i \neq j : \|c_i - c_j\| < 24$ px
Zone occupancy	$\#\{\text{persons in ROI}\} >$ capacity
Count rules	$\#\{c\} \square n, \square \in \{=, \geq, \leq, >, <\}$
Time-of-day	$t \in [t_{\text{start}}, t_{\text{end}}]$ (cross-midnight)

spec carries no separate cadence field). In this submission the NL-to-rule materializer is a deterministic keyword/regex module (`alert_materializer.py`: creation/retrieval intents, event/time keywords, τ/cooldown numbers, clarify/abstain classification), not LLM structured-output; the LLM is used only for query parsing.

C. Authoring example

The utterance *"alert me when more than five people gather in Zone A after 10 PM"* produces

$$\mathcal{R} = \langle \text{crowd},\ \{\text{person}\},\ \text{Zone A},\ 5,\ [22{:}00, 06{:}00],\ \text{med},\ 60,\ \{\text{snapshot}, \text{push_ws}\},\ 1\text{Hz} \rangle$$

without any manual configuration, where $[22{:}00, 06{:}00]$ applies the default 06:00 end when no end time is specified (night $\rightarrow$ 20:00–06:00 policy). Six rule families are evaluated per-frame against the live detection stream (Table VI); each rule is cached from MongoDB with a 5-second TTL. Fight fires on $\bar{\mathcal{E}}_W \geq 0.03$ with person-center distance $<$ 24 px ($\approx$ overlapping persons at 1080p). Unknown zones, missing thresholds, and unsupported verbs are guarded by schema validation that requests clarification instead of materializing a rule; ambiguous cases (fuzzy zone, implicit "a few", cross-midnight) account for the 9% non-perfect rules in §VIII.

D. Validation safeguards (clarify/abstain)

Unknown zones, missing thresholds or time windows, and unsupported verbs (e.g., "follow him home", "identify the person") are guarded by schema validation that requests clarification instead of materializing a rule. The benchmark release contains 120 alert utterances (60 clean + 60 stress: ambiguous, conflicting, missing, unsupported); the reported 91% rule-correctness is on the 60 clean utterances, while the stress set exercises the clarify/abstain path implemented in `backend/app/services/alert_validator.py`. Classification of stress kinds and validation of the 9-tuple rule are deterministic and tested (`backend/tests/test_alerts.py`).

E. Complementary statistical anomaly detector

To surface unknown behaviors we maintain a rolling 7-day baseline of detection counts $\{x_{c,h}^{(d)}\}_{d=1}^{7}$ per (camera, hour) bucket and emit an event when

$$Z_{c,h} = \frac{x_{c,h}^{*} - \mu_{c,h}}{\sigma_{c,h} + \varepsilon} \geq \tau_Z, \tag{7}$$

Algorithm 1 Detection and indexing per frame.

Input: frame F_t, timestamp t
$D_t \leftarrow \text{YOLOv11mSeg}(F_t, \tau_{\text{det}}{=}0.5)$
for each $(b_i, m_i, c_i, p_i) \in D_t$ **do**
  $\mathbf{q} \leftarrow \text{LABHistogramMode}(F_t[b_i], m_i)$
  $C \leftarrow \arg\min_{\mathbf{p}_j} \Delta E_{00}(\mathbf{q}, \mathbf{p}_j)$
  if $c_i{=}\texttt{person} \wedge w(b_i) \geq 60$ **then**
    $\hat{y} \leftarrow \sigma(\text{PAR}(\text{Letterbox}(\text{Mask}, 80, 160)))$
    $\mathbf{e} \leftarrow$ Eq. (6); $\text{FAISS}_p.\text{add}(\mathbf{e})$
  end if
  MongoDB.insert(record)
end for
if $t \bmod \Delta_{\text{idx}} = 0$ **then**
  EnqueueSemanticIndexing(F_t)
end if

with $\tau_Z \in \{1.5, 2.0, 3.0\}$ mapping to $\{\text{low}, \text{medium}, \text{high}\}$ for unusual-object, off-hours, and crowd-spike events; $+\varepsilon$ guards $\sigma{=}0$. Unusual-object uses a count heuristic, off-hours fires when no baseline exists, and crowd-spike uses $z{\geq}2.0$. On a 7-day labeled subset the detector reaches precision/recall/F_1 of $0.872/0.811/0.840$ at 1.4 FP/day (crowd-spike), $0.804/0.866/0.834$ at 2.1 FP/day (off-hours) and $0.713/0.792/0.751$ at 3.7 FP/day (unusual-object); the last false-positive rate is acknowledged high.

VII. Implementation Detail (Brief)

The complete detection-and-indexing loop is given in Algorithm 1. The end-to-end pipeline is organized as five dataflow layers (capture, perception, indexing, reasoning, output) in which the three novel components (C1, C2, C3) are the only non-off-the-shelf blocks.

Query resolution. The NL query q is parsed by an LLM into a typed filter F, a semantic query q_{sem} and an intent ι; structured candidates are fetched as $R_s \leftarrow \text{MongoDB.find}(F)$ and semantic candidates as $R_v \leftarrow \text{FAISS.search}(g_T(q_{\text{sem}}), k)$, where g_T is OpenCLIP text for scene queries and MiniLM with $\mathbf{v}{=}\mathbf{0}$ for person queries against the 1152-d index. The mixing weight is $\alpha \leftarrow \alpha(F, \iota, |R_s|, |R_v|)$ (Eq. (3); action queries skip R_s); R_s and R_v are merged and rescored by Eq. (1) on per-query normalized scores, then passed through a confidence filter, the recency boost $(1{+}0.5{\cdot}2^{-age/24h})$ and MMR ($\lambda{=}0.3$, unablated), and the top-k clips plus an LLM answer are returned.

VIII. Evaluation

A. Setup

Hardware: Intel i7-13700K, RTX 4070 (12 GB), 32 GB DDR5, Windows 11. Software: Python 3.12, PyTorch 2.3, CUDA 12.1, OpenVINO 2024.2, FAISS 1.8, MongoDB 7. **Full deployment.** Four 1080p IP cameras at 2 FPS per stream for $\sim$8 h ($\sim$230k frames, 1.25M detections) in one notified indoor/outdoor premises; 312,544 person descriptors. **Deployment retrieval benchmark.** A 150-query NL benchmark (30 each: count, find, track, action, filter) authored

from live-clip observations, with human-annotated relevant clip sets; for counting, relevance means the clip contains the true count and P@5 measures supporting-clip retrieval, with exact-count EM/MAE complementary. Its per-query export, annotator roster and tune/test partition were not preserved, so the deployment CIs below are query-level and the deployment numbers are the recorded measurements of that study. **Released reproducibility track.** The release ships a single-camera subset (123 clips, 640×480, 5 FPS, 3.6 s each, ∼7.2 min inside a 2 h window; 242 scene frames; 1,778 person descriptors) with metadata-derived rather than human labels, a 50/100 tune/test split, 120 alert utterances (30/30 dev/test) and 2,016 track-disjoint person pairs (528 calibration); the harness is in `backend/scripts/eval_*.py` and the outputs in `results/`. The deployment study used 60 held-out utterances with reference 9-field rules and a 50-pair calibration split (no CV, gallery disjointness not logged). Three encoders are used by role: OpenCLIP ViT-B/32 (scene text), SigLIP-base (person visual), MiniLM-L12-v2 (attribute text); no encoder ablation is claimed.

Statistical protocol. We report mean P@5 with a 95% bootstrap CI (10,000 query-level resamples); macro-mean is the mean of the five category means (n=30 each). Paired comparisons (adaptive vs. fixed-α, concat vs. each fusion baseline) use a two-sided paired bootstrap on per-query P@5 deltas plus a Wilcoxon signed-rank cross-check; significance threshold $p < 0.05$ without multiple-testing correction (find p=0.04 and track p=0.07 are marginal and do not survive Holm correction for five categories). The deployment per-query export was not preserved, so grouped resampling by video/identity could not be recomputed for it; the released-subset track reports a grouped bootstrap over video/time groups together with Holm-corrected per-category tests (3,000 resamples) from `backend/app/eval/stats.py`, and correlated queries from the same video may narrow CIs.

B. Evaluation protocol (definitions and thresholds)

Precision@5 is $|\text{top}{-}5 \cap \text{relevant}|/5$; Recall@10 is $|\text{top}{-}10 \cap \text{relevant}|/|\text{relevant}|$; the paper's F_1@10 is the harmonic mean of P@5 and R@10 (not the standard P@10-based F_1). Macro-mean is the unweighted mean of the five category means (30 queries each). For counting queries we additionally report exact-match (EM) accuracy and mean absolute error (MAE) on the predicted count; P@5 for counting measures retrieval of supporting clips that contain the true count. Thresholds used throughout: detector confidence τ_{det}=0.5; fight trigger $\bar{\mathcal{E}}_W \geq 0.03$ with person-center distance < 24 px; Z-score $\tau_Z \in \{1.5, 2.0, 3.0\}$ for $\{\text{low}, \text{medium}, \text{high}\}$ anomaly severity. The stochastic same-space projection baseline is rerun with multiple seeds on the released subset (mean±SD reported there); grouped bootstrap by video/time group and Holm correction for the five categories are implemented in `backend/app/eval/stats.py` and applied to every released-subset comparison, while the deployment table keeps the recorded single-run measurements.

TABLE VII
PER-CATEGORY P@5 WITH 95% BOOTSTRAP CIS (10,000 RESAMPLES, n=30 PER CATEGORY, n=150 OVERALL). Δ IS THE PER-QUERY MEAN OF (ADAPTIVE − PER-CATEGORY BEST-FIXED-α); p IS TWO-SIDED PAIRED BOOTSTRAP. THE BOTTOM ROW COMPARES THE BEST SINGLE FIXED-α MACRO-MEAN (α=0.3) AGAINST ADAPTIVE-α.

Category	Best fixed-α	Adaptive	Δ (p)
Count	0.94 ± 0.05 (α=1.0)	0.94 ± 0.05	+0.00 (n.s.)
Find	0.76 ± 0.07 (α=0.3)	0.78 ± 0.07	+0.02 (0.04)
Track	0.80 ± 0.06 (α=0.3)	0.81 ± 0.06	+0.01 (0.07)
Action	0.69 ± 0.08 (α=0.0)	0.69 ± 0.08	+0.00 (n.s.)
Filter	0.92 ± 0.05 (α=1.0)	0.92 ± 0.05	+0.00 (n.s.)
Macro-mean (best single fixed-α)	0.666 ± 0.039	$\mathbf{0.828 \pm 0.034}$	$\mathbf{+0.162}$ ($< 10^{-4}$)

C. C1 – adaptive-α vs. fixed-α

Adaptive-α raises macro-mean P@5 from the best fixed-α value $0.666\,(\pm 0.039)$ at α=0.3 to $0.828\,(\pm 0.034)$, an absolute gain of $+0.162$ with paired-bootstrap 95% CI $[+0.118, +0.207]$ over 150 queries (Table VII). A per-intent fixed-α lookup derived from Table II (count→1.0, action→0.0, filter→1.0, find/track→0.3) reaches 0.822 macro-mean ($0.94{+}0.76{+}0.80{+}0.69{+}0.92/5$), so the soft terms ($\rho_{\text{cplx}}, \rho_{\text{src}}$, $0.6/0.2/0.2$ unablated) contribute $+0.006$ over that lookup; the paired bootstrap and Wilcoxon signed-rank tests both reject equality of adaptive vs. best single fixed-α at $p < 10^{-4}$; per-category gains against the per-category best fixed-α are zero on the override categories (count, action) and on filter (where α=1.0 already saturates), and positive on find ($+0.02$, $p = 0.04$) and track ($+0.01$, $p = 0.07$, both marginal without correction). Crucially, the gain in macro-mean comes from the fact that no single fixed α can match the count, action, and filter optima simultaneously (Observation 1); adaptive-α does so by construction.

a) Released-subset check (C1).: The same code path was re-run on the released single-camera subset with a protocol-matched control (50 tune / 100 test queries, metadata-derived labels, `backend/scripts/eval_retrieval_control.py`). Adaptive-α reaches macro P@5 0.538, identical to the best tune-selected fixed α (0.538 at α=0.7; Δ=0.000, p=1.0) and above the paper's fixed per-intent lookup (0.428), while the extreme fixed values collapse (0.048 at α=0.0, 0.380 at α=0.3). Per category, adaptive-α is ahead wherever the overrides fire (count 0.390 vs. 0.010 at α=0.0; action 0.090 vs. 0.000 at α=1.0), which reproduces the override-gap observation of Table II; the magnitude of the deployment gain is not reproduced on the subset, whose labels are metadata-derived over a single 2-hour window, and we do not claim that it is.

b) Competitors and robustness (C1).: Three further comparisons run on the same control. (i) A learned competitor: a multinomial logistic model that predicts the best fixed α per query from the intent ι, the branch sizes n_s and n_v and the filter size $|F|$, fitted on the 50 tune queries and evaluated on the 100 test queries, reaches 0.534, below the 0.538 of the training-free rule: its oracle-best-α label is 0.5 on 49 of the 50 tune queries, so the fitted model collapses to a constant α=0.5

and loses exactly the override cases that the closed-form rule switches on. (ii) Soft-weight sensitivity: over all 16 triples of the three soft weights of Eq. (4) that sum to one, searched on the tune split, the test macro P@5 is 0.538 for every triple, so the released result does not hinge on the hand-set $0.6/0.2/0.2$; on the subset the two hard overrides carry the whole effect (soft-term contribution $+0.000$, against $+0.006$ on the deployment). (iii) Counting is label-limited here: the window-mode caption estimate reaches EM 0.10/MAE 2.20 against the anchor-frame caption count, while caption counts inside one window span 4.5 persons on average, so no window-based predictor can beat the labels themselves (`backend/scripts/eval_alpha_competitors.py`).

D. C2 – fusion ablation

The heterogeneous-space concat descriptor reaches $F_1@10 = 0.807\,(\pm 0.024)$, beating the best unimodal index (0.700 ± 0.031) by $+10.7$ pp (paired bootstrap 95% CI $[+7.9, +13.4]$, $p < 10^{-4}$) and the best fusion baseline RRF (0.785 ± 0.026) by $+2.2$ pp (95% CI $[+0.4, +4.1]$, $p = 0.018$, Wilcoxon $p = 0.022$). The same-space additive baseline (single-seed random Gaussian projection) gives $+5.4$ pp over visual-only ($p = 0.003$) but remains 5.3 pp below the concat construction ($p = 0.001$). Significance therefore holds against both unimodal and fusion baselines, with the concat-vs-RRF margin being the smallest and least conservative; a gallery-weight sweep on the same deployment set gives $F_1@10$ 0.785 at $(\alpha_v, \beta_t)=(0.5, 0.5)$, 0.799 at $(0.7, 0.3)$ and 0.774 at $(0.4, 0.6)$, so the RRF tie is a property of that weight rather than of the fusion family. Multi-seed means for the stochastic same-space projection baseline (five seeds, P@5 0.436 ± 0.0004) and a learned-projection competitor are measured on the released subset, whose 312k index and per-query export are not released (`backend/app/eval/baselines.py`, `backend/scripts/eval_fusion.py`). *Released-subset scope.* The subset carries pseudo-tracks rather than identity labels, so it cannot test the deployment fusion result; that evaluation is published with the release (`results/fusion_tables.md`) as a code-path check, and Table IV remains the evidence for C2.

E. C3 — alert authoring

The 60 held-out utterances exercise all nine rule fields (Table VIII); "Rule-correctness" requires all nine to be correct. Closed-vocabulary defaults (severity `info`, cadence 1Hz) reach 1.00 when unspecified; conditional accuracy when specified is lower. The hardest specified fields are zone naming (0.88, fuzzy references such as *"the back entrance"* vs. canonical `Zone B`) and time window (0.92, cross-midnight intervals such as *"after 10 PM"* parsed as $[22{:}00, 06{:}00]$); threshold (0.93) misses on implicit quantifiers (*"a few"*), and object-set expansion (0.95) under-expands category nouns (*"vehicle"* $\rightarrow$ $\{$`car`$\}$). Aggregated, 0.91 of utterances yield a fully-correct rule on the first attempt; the remaining 9% require a single zone- or threshold-level correction, never a structural rewrite.

TABLE VIII
CONVERSATIONAL ALERT AUTHORING ON A HELD-OUT 60-UTTERANCE SET. "FIELD-LEVEL" = PER-FIELD EXACT MATCH, "RULE-CORRECTNESS" = ALL 9 FIELDS CORRECT.

Field	Acc.	Hardest mistake
Event family	0.97	loiter vs. zone
Object set	0.95	"vehicle" $\rightarrow$ {car, truck}
Zone	0.88	fuzzy zone names
Threshold τ	0.93	implicit ("a few")
Time window	0.92	cross-midnight
Severity	1.00	—
Cooldown	0.98	—
Side-effects	0.95	implied snapshot
Cadence	1.00	—
Rule-correctness	**0.91**	—

TABLE IX
SUBSTRATE METRICS (YOLOV11M-SEG DETECTION, BOT-SORT+OSNET TRACKING MEAN OVER 4 CAMERAS, PAR MACRO, COLOR NAMING ON 1,568 ROIS, Z-SCORE ANOMALY).

Stage	Metric	Value
Detection (person)	Precision / F_1 / mAP@0.5	0.965 / 0.927 / 0.873
Tracking (mean)	MOTA / IDF1	0.909 / 0.865
Person attributes	Macro F_1 (7 attrs)	0.882
Color naming	CIEDE2000 / RGB-Euclid	0.923 / 0.821
NL parsing	Field-correctness	0.94

TABLE X
END-TO-END QUERY LATENCY BY CATEGORY. "SEM. ONLY" ISOLATES THE FAISS LOOKUP (SCENE-INDEX MEAN 108.6MS); "TOTAL" INCLUDES LLM PARSING, HYBRID BLENDING, MMR, AND CLIP STITCHING. PERSON INDEX HAS N=312,544 DESCRIPTORS WITH 109MS MEDIAN; SCENE AND PERSON NUMBERS ARE NOT INTERCHANGEABLE. TOTAL MEAN IS 6.0S (FIND 12.0S, PARSE OUTLIER 6.4S ON FIND); FAISS IS NOT THE BOTTLENECK.

Category	Parse (ms)	Sem. only (ms)	Total (ms)	P@5
Count	1,768	94	3,441	0.94
Track	2,101	109	5,413	0.81
Find	6,405	121	11,983	0.78
Action	2,423	137	6,131	0.69
Filter	1,590	82	3,108	0.92
Mean	2,857	**108.6**	6,015	**0.83**

a) Released alert benchmark.: The release ships 120 author-constructed utterances (60 clean; 60 stress covering ambiguous, conflicting, missing and unsupported requests) with a 30/30 dev/test split. With the deterministic materializer and the clarify/abstain validator, all 30 clean *test* utterances yield nine exactly correct fields and all 30 stress *test* utterances are clarified with zero false materializations. Being author-constructed over a closed vocabulary, this is a coverage/regression measurement, not the benchmark behind the 0.91 of Table VIII; the two are reported separately.

F. Supporting numbers

Substrate metrics are summarized in Table IX; the Z-score anomaly detector (Eq. (7)) is reported in the accompanying text.

The 4-camera deployment sustains 2.0 FPS per stream for detection/tracking at 4.2 GB resident memory, 7.8 GB VRAM, and 68% mean GPU utilization (91% p95) on the RTX 4070; VLM captioning (Qwen2-VL-2B with fallbacks) runs on keyframes only and is excluded from the 2-FPS tracking budget. End-to-end NL query latency is interactive-only (mean 6.0s, Find 12.0s incl. 6.4s parse outlier; per-category means in Table X). `backend/scripts/measure_latency.py` profiles the released subset (CPU, 242 scene and 1,778 person vectors): the scene retrieval core is 37.4ms mean / 37.4ms p50 / 39.7ms p95 and the person retrieval core 17.2ms mean / 18.5ms p50 / 20.4ms p95, and 1–8 concurrent workers sustain 26.5–39.5 queries/s at p95 job latency 39.6–265.4ms. These component figures exclude LLM parsing, clip stitching, MongoDB and ingestion, and are not 312k-descriptor deployment percentiles; the 108.6ms FAISS mean of the deployment is the retrieval step alone and is not the bottleneck.

IX. Discussion and Conclusion

Generalization. All three contributions are schema-decoupled: C1 needs only a category function $C(\cdot)$ with finitely many overrides; C2 admits any pair of frozen encoders, with only (α_v, β_t) learned on a 50-pair calibration set; C3's grammar requires only a typed-schema and trigger-function extension per new event family. Evaluation here is a single 4-camera premises ($\sim$8 h); cross-site, night/rain, and viewpoint-shift evaluation is follow-up.

Limitations. (i) Observation 1 is empirical: it assumes steep degradation away from per-category optima on our benchmark and is not guaranteed in general. Crowded scenes, partial occlusion, camera-domain shifts and ambiguous action phrasing degrade both branches and are not isolated here. (ii) Concat fusion costs $d_v + d_t$ bytes per descriptor (≈ 1.44 GB for 312k flat); galleries with $N > 10^7$ would motivate IVF-PQ over flat IP; MMR (λ=0.3) and recency are unablated. (iii) Alert-grammar correctness depends on the extraction path: the deployment study used an LLM extractor (a 6-pp drop with local 7B models), whereas the release ships a deterministic keyword/regex materializer that routes ambiguous zones, implicit thresholds ("a few"), cross-midnight windows and unsupported commands to a clarify/abstain validator (30/30 clarifications, no false materializations). The paths are not interchangeable, and the released coverage measurement is not an open-world accuracy estimate; no OSNet-only or public-ReID baseline is reported. (iv) *Demographic bias, privacy and consent.* PAR-crossroad-0238 was trained on a corpus skewed toward adult, daylight imagery, so its per-attribute accuracy degrades on children, occluded subjects, low-light scenes and under-represented clothing styles; its binary *gender* attribute is a coarse visual proxy, and no demographic-bias evaluation is possible without demographic labels. Live capture was in notified premises with face imagery retained per-camera for at most 8 h. The release implements the corresponding controls in code—opt-in role/camera/zone access control with audit logging, an explicit 8 h retention prune, a face-blur utility and a gender-off output filter—disabled by default so that the bundled demo is unchanged; they need an operator and a trusted reverse proxy, do not rebuild the shipped index, and the face-blur utility is not wired into live ingestion. We mitigate by using gender only as a soft cue; no inter-annotator agreement, cross-camera generalisation or demographic-bias measurement is claimed.

Conclusion. The three contributions raise macro-mean P@5 from 0.67 to 0.83, gain 10.7-/2.2-pp F_1@10 over uni-modal/RRF baselines, and reach 91% full-rule alert-authoring correctness on a 4-camera, 312k-descriptor deployment, showing that schema-aware fusion closes the gap between dense retrieval and a usable conversational surveillance interface. The release adds query/relevance definitions, eval scripts, fusion parameters, alert schema, the privacy/access switches and a runnable single-camera reproducibility track with measured outputs, so that the deployment numbers can be independently checked and the code paths can be exercised end-to-end.

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